

Welcome to MATH221 Linear Algebra I!

Instructor: Changho Han (한창호)

Email: changho_han@korea.ac.kr

Office Hour: TBA

Grade Distribution:

10% Attendance (10 - # absence, $0 = F$) ← University Policy

20% Quizzes (2 Quizzes)

35% Midterm

35% Final Exam

More info
in LMS

Expectations in the classroom:

- Lectures are in ENGLISH.
- Feel FREE to ASK Questions anytime during the class (you are encouraged to do so). NO

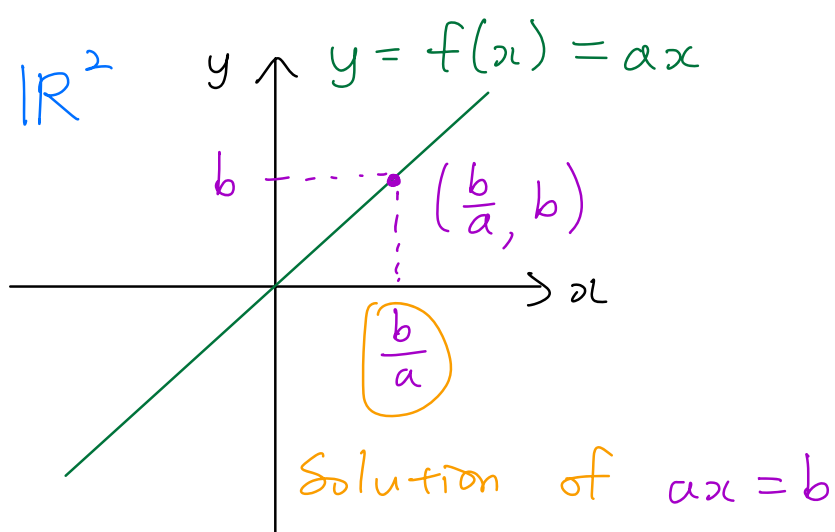
Questions are Dumb: it helps others.

- But do your best to ask in English! I will translate Questions (in Korean) into English.

So what is Linear Algebra?

Study of vector spaces and higher dimensional analog
of linear functions & equations. $(\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3, \text{etc})$

$\uparrow f(x) = ax$ $\uparrow ax = b$



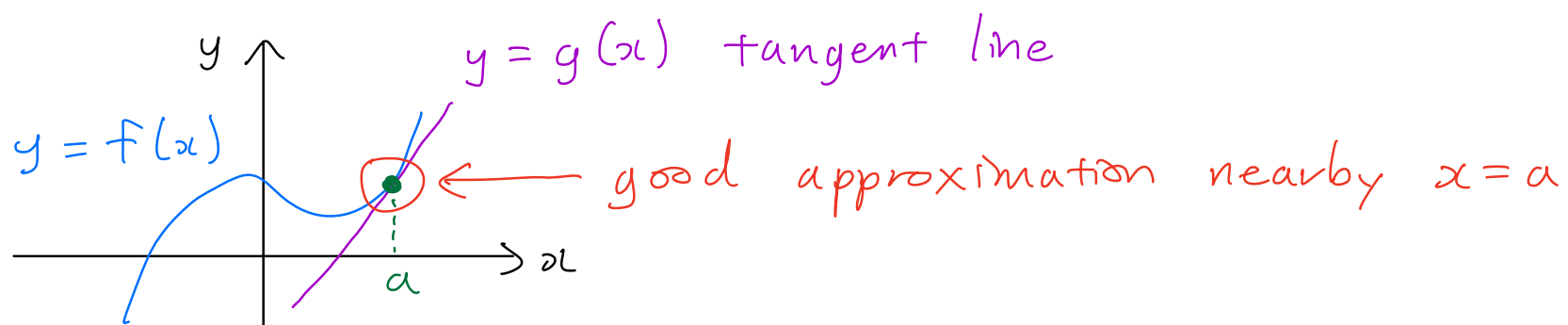
Here, a Vector refers to a point of a vector space.

For example,

- Science: an arrow $\left(\nearrow \right)$
- AI/CS: pixelated image $\left(\begin{array}{|c|c|} \hline \text{red} & \text{blue} \\ \hline \text{green} & \text{yellow} \\ \hline \end{array} \right)$
- Stats: a row of dataset $([0.7, 5.2, 4.9, 10, 8.1])$

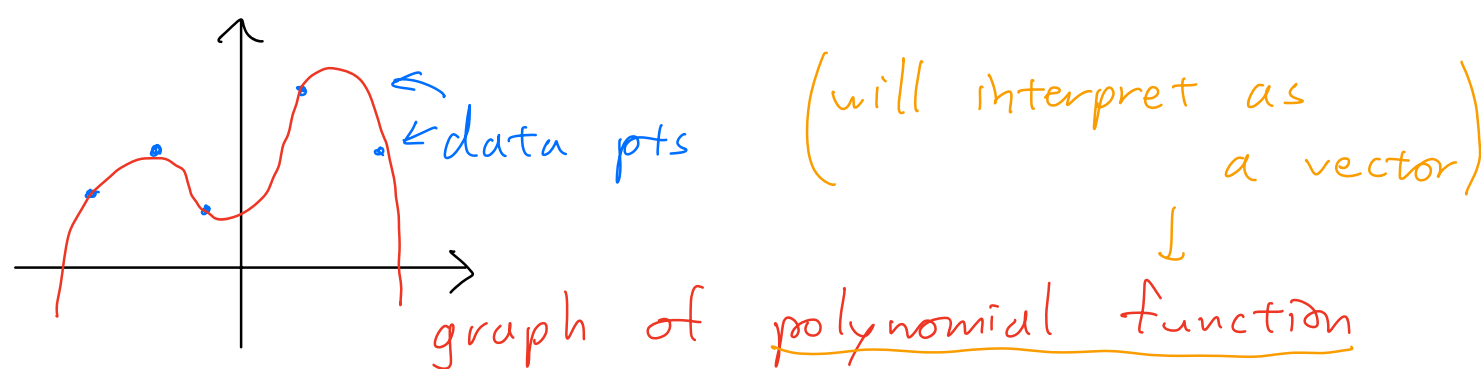
Why Linear Algebra?

- Linear Algebra, just like Calculus, is a building block of modern math.
- Linear Functions $\sim$ Complicated Functions

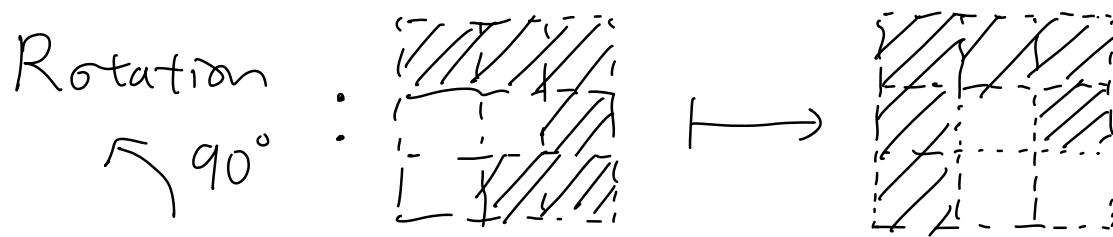


• Traditional Applications

◇ Lagrange Interpolation: find a polynomial function passing through n data points in $\mathbb{R}^2$



◇ Image Manipulation: by interpreting a pixelated image as a vector, stretch/squeeze/shear/rotate that image (via linear map)

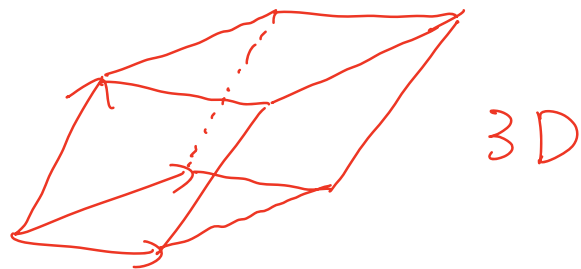
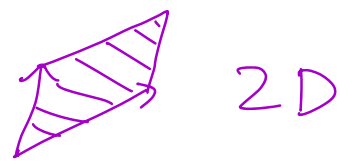


◇ Solving Systems of Linear Equations

(Algorithmically): Solve something like

$$\begin{array}{l|l} 4x + 3y = 9 & x + y + z = 4 \\ x + y = 2 & x + 2y = 3 \\ & -y + z = 1 \end{array}$$

◇ (Signed) Volume of Parallelograms and Parallelepipeds



(via determinants)

This hybrid theory/computation course is about:

- Fundamentals of Linear Algebra
 - Starts with vector space instead of systems of linear equations
 - Build a proper foundation to higher math
 - Combines repetitions/patterns into a single package
- Proof-based introduction
 - Why is statement X true? Prove it!
- Course Structure:
 - (1) First 8 weeks: Heavy Theory - Building
 - (2) Last 8 weeks: Computational Applications
(with some theory)

- Unit Notes: The main textbook is used only for Practice Problems. For the content of this course, Unit Notes will be used instead. Those will be up in LMS as pdf files.

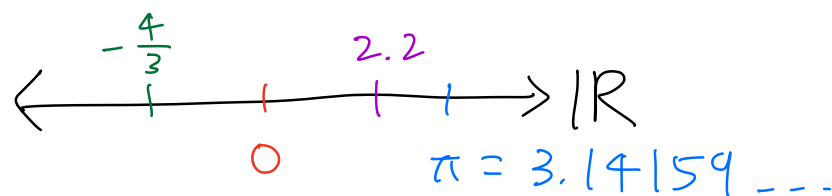
Unit 1: Vector Spaces

Goal: Define vectors & vector spaces, then study their properties (so build foundations)

§1.1 $\mathbb{R}^n$ as a Vector Space

Before jumping into $\mathbb{R}^n$, let's start with $\mathbb{R}^2$.

Here, $\mathbb{R}$ means the set of all real numbers



Definition

Def 1.1: $\mathbb{R}^2 \stackrel{\text{LHS}}{:=} \{ \underbrace{(a, b)}_{\text{a pair}} : \underbrace{a, b \in \mathbb{R}}_{\text{means } a \text{ \& } b \text{ are elements of } \mathbb{R}} \}$

(Left-hand side is defined by Right-hand side)

Note: So $\mathbb{R}^2$ is a set of pairs of real numbers.
2-tuples $\nearrow$

Cartesian plane is a
way to visualize $\mathbb{R}^2$

